



Week 01 – Network Security Lab Foundations

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1. Objective

To initialize a professional hybrid laboratory environment and master the mechanics of low-level network communication by building an isolated, controlled virtual network suitable for executing high-risk security simulations safely.

2. Lab Environment Overview

2.1 System Specifications

Component	Attacker (Kali)	Victim (Windows 10)
Operating System	Kali Linux	Windows 10 Pro
RAM	4 GB	2 GB
Processors	4	2
Storage	50 GB (NVMe)	20 GB (SCSI)
Network Adapter	Custom vmnet2	Custom vmnet2
IP Address	192.168.100.130	192.168.100.128
Subnet Mask	255.255.255.0	255.255.255.0

2.2 Virtualization Platform

- **Software:** VMware Workstation Pro
- **Network Type:** Custom Virtual Network (vmnet2) – fully isolated
- **Purpose:** Prevent leakage while allowing inter-VM communication



3. Task 01: Virtualization & Network Topology Setup

3.1 Description

Build the foundation of the Purple Team lab by configuring isolated attack and defense segments using VMware Workstation. Both VMs were assigned to a custom internal network (vmnet2) for an air-gapped testing environment.

3.2 VMware Workstation – Both VMs Running

Both virtual machines (kali-linux and Win10) are visible and running in VMware Workstation.

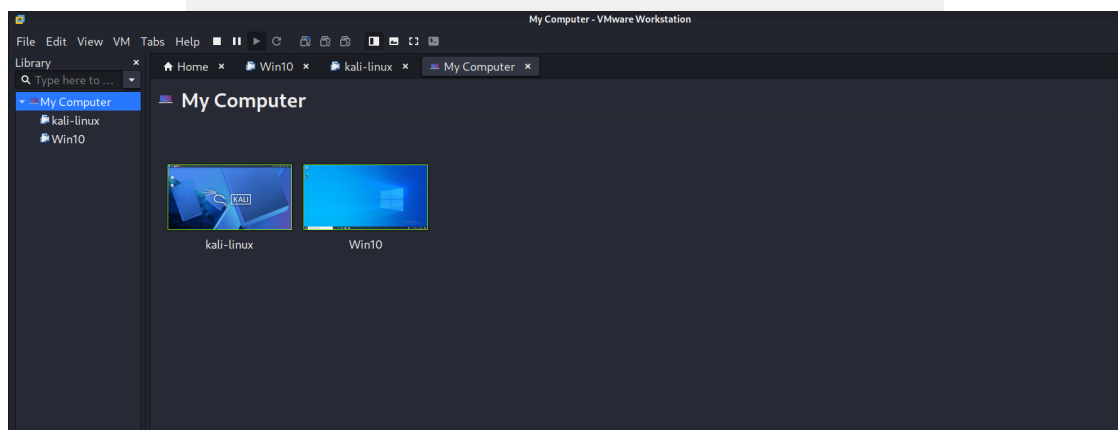


Figure 1: VMware Workstation – kali-linux and Win10 VMs

3.3 Network Adapter – Kali Linux

Kali Linux was configured with Custom (vmnet2) adapter. Specs: 4 GB RAM, 4 CPUs, 50 GB NVMe.

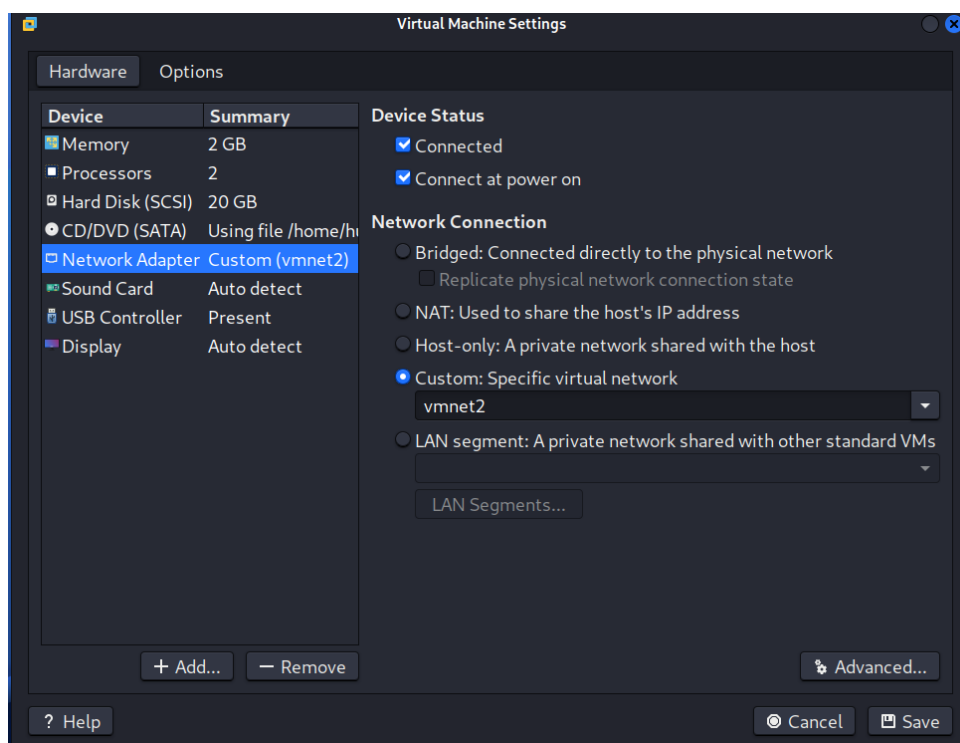


Figure 2: Kali Linux VM Settings – Custom vmnet2 network

3.4 Network Adapter – Windows 10

Windows 10 victim machine similarly configured on Custom (vmnet2). Specs: 2 GB RAM, 2 CPUs, 20 GB.

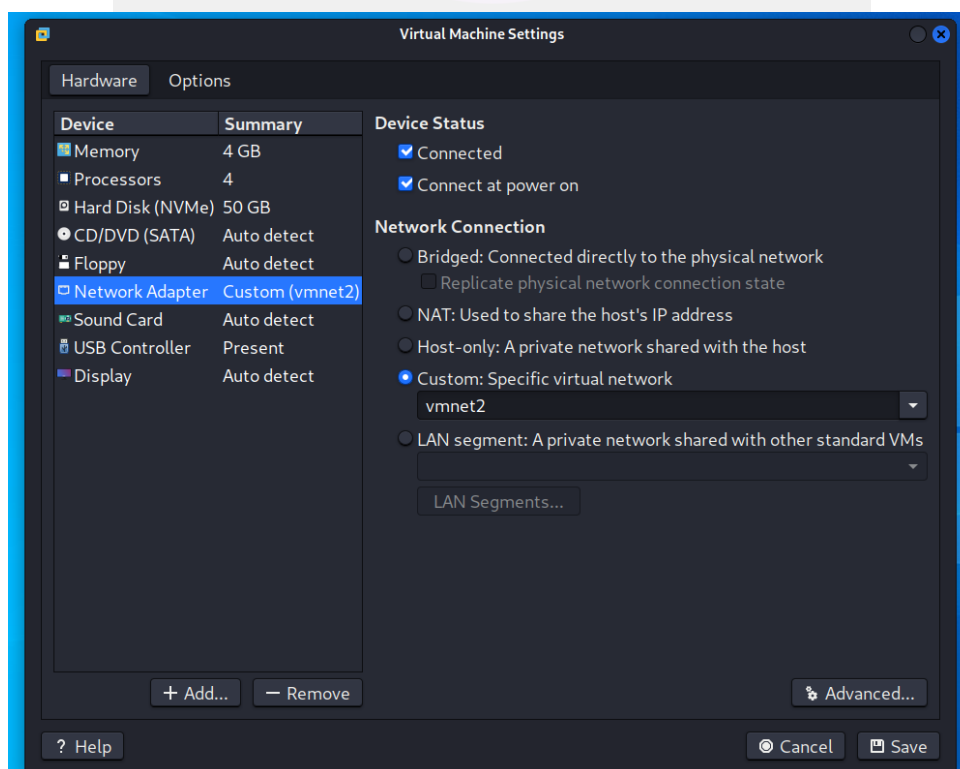




Figure 3: Windows 10 VM Settings – Custom vmnet2 network

4. Static IP Assignment

4.1 Kali Linux – ip a Output

Static IP verified using `ip a` command. `eth0` assigned **192.168.100.130/24**.

```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)~  
$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def  
ault qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g  
roup default qlen 1000  
    link/ether 00:0c:29:dd:90:45 brd ff:ff:ff:ff:ff:ff  
    inet 192.168.100.130/24 brd 192.168.100.255 scope global dynamic noprefix  
route eth0  
        valid_lft 1723sec preferred_lft 1723sec  
    inet6 fe80::20c:29ff:fedd:9045/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
(kali@kali)~  
$
```

Figure 4: Kali Linux `ip a` output – 192.168.100.130

Interface:	eth0
IPv4 Address:	192.168.100.130
Subnet Mask:	255.255.255.0 (/24)
MAC Address:	00:0c:29:dd:90:45

4.2 Windows 10 – ipconfig Output

`ipconfig` confirmed static IP **192.168.100.128/24** on Ethernet0.



```
Command Prompt
Microsoft Windows [Version 10.0.19045.2965]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Muhammad Hussain>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : localdomain
    Link-local IPv6 Address . . . . . : fe80::c3e0:14e0:68eb:4c97%4
    IPv4 Address. . . . . : 192.168.100.128
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

C:\Users\Muhammad Hussain>
```

Figure 5: Windows 10 ipconfig output – 192.168.100.128

Adapter:	Ethernet0
IPv4 Address:	192.168.100.128
Subnet Mask:	255.255.255.0
Default Gateway:	(blank – isolated network)

5. Connectivity Testing

5.1 Ping: Kali Linux to Windows 10

Ping from Kali (192.168.100.130) to Windows (192.168.100.128) – 4/4 packets received, 0% loss.

```
kali@kali: ~
Session Actions Edit View Help
┌─(kali@kali)-[~]
│$ ping 192.168.100.128
│PING 192.168.100.128 (192.168.100.128) 56(84) bytes of data:
│64 bytes from 192.168.100.128: icmp_seq=1 ttl=128 time=1.51 ms
│64 bytes from 192.168.100.128: icmp_seq=2 ttl=128 time=1.44 ms
│64 bytes from 192.168.100.128: icmp_seq=3 ttl=128 time=1.59 ms
│64 bytes from 192.168.100.128: icmp_seq=4 ttl=128 time=1.46 ms
│^C
│— 192.168.100.128 ping statistics —
│4 packets transmitted, 4 received, 0% packet loss, time 3006ms
│rtt min/avg/max/mdev = 1.440/1.499/1.585/0.055 ms
│
└─(kali@kali)-[~]
  $
```



Figure 6: Ping from Kali to Windows – successful

```
$ ping 192.168.100.128
PING 192.168.100.128 56(84) bytes of data.
64 bytes from 192.168.100.128: icmp_seq=1 ttl=128 time=1.51 ms
64 bytes from 192.168.100.128: icmp_seq=2 ttl=128 time=1.44 ms
64 bytes from 192.168.100.128: icmp_seq=3 ttl=128 time=1.59 ms
64 bytes from 192.168.100.128: icmp_seq=4 ttl=128 time=1.46 ms
4 packets transmitted, 4 received, 0% packet loss
rtt min/avg/max/mdev = 1.440/1.499/1.585/0.055 ms
```

5.2 Ping: Windows 10 to Kali Linux

Reverse ping from Windows (192.168.100.128) to Kali (192.168.100.130) also successful.

```
Home x Win10 x kali-linux x
Select Command Prompt
Microsoft Windows [Version 10.0.19045.2965]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Muhammad Hussain>ping 192.168.100.130

Pinging 192.168.100.130 with 32 bytes of data:
Reply from 192.168.100.130: bytes=32 time<1ms TTL=64
Reply from 192.168.100.130: bytes=32 time=1ms TTL=64
Reply from 192.168.100.130: bytes=32 time=1ms TTL=64
Reply from 192.168.100.130: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.100.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\Muhammad Hussain>
```

Figure 7: Ping from Windows to Kali – successful

```
C:\Users\Muhammad Hussain>ping 192.168.100.130
Reply from 192.168.100.130: bytes=32 time<1ms TTL=64
Reply from 192.168.100.130: bytes=32 time=1ms TTL=64
Reply from 192.168.100.130: bytes=32 time=1ms TTL=64
Reply from 192.168.100.130: bytes=32 time=1ms TTL=64
Packets: Sent=4, Received=4, Lost=0 (0% loss)
Minimum=0ms, Maximum=1ms, Average=0ms
```

6. Results Summary



Task	Status	Details
VMware Workstation Setup	DONE	2 VMs created
Kali Linux Installed	DONE	4GB RAM, 4 CPU
Windows 10 Installed	DONE	2GB RAM, 2 CPU
Custom vmnet2 Network	DONE	Both VMs on vmnet2
Kali Static IP	DONE	192.168.100.130/24
Windows Static IP	DONE	192.168.100.128/24
Kali to Windows Ping	PASS	0% packet loss
Windows to Kali Ping	PASS	0% packet loss

7. Learning Outcomes



1. Mastered installation and configuration of VMware Workstation for a dual-VM cybersecurity lab environment.
2. Understood the importance of using Custom Internal Network (vmnet2) to completely isolate the lab from the host's physical network.
3. Successfully assigned and verified static IPs on Linux (`ip a`) and Windows (`ipconfig`).
4. Validated inter-VM connectivity using ICMP ping in both directions, confirming a stable, controlled sandbox for executing high-risk security simulations.
5. Gained foundational knowledge essential for upcoming Purple Team exercises involving attack simulations, packet analysis, and incident response.

CyberStep | Purple Team Internship – Week 01 Lab Report

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